HAMZA MOUNIR

SENIOR SOFTWARE ENGINEER | SCALABLE SYSTEMS | CLOUD ARCHITECTURE | AI SYSTEMS | TECHNICAL LEADERSHIP

LANGUES
FRENCH (NATIVE)
ENGLISH (PROFESSIONAL)

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Senior Software Engineer with extensive experience designing scalable platforms, microservice architectures, and cloud-native systems. I specialize in building reliable infrastructures that allow companies to scale from early startup stages to enterprise-grade platforms.

My expertise spans backend architecture, distributed systems, DevOps automation, and AI-powered applications. I build systems that are resilient, automated, and operationally efficient, reducing friction for both engineering teams and business operations.

On the frontend, I deliver high-performance web applications using React and Next.js, focusing on user experience, accessibility, and seamless integration with complex backend services and AI-driven systems.

I also design and deploy AI-powered solutions, including LLM-based services, retrieval-augmented generation (RAG), vector search, and context-aware systems. I build infrastructure to optimize token usage, manage streaming responses, and integrate intelligent automation into production environments.

Beyond coding, I actively improve engineering processes, development workflows, and team collaboration, mentoring engineers and ensuring teams can deliver faster with higher reliability.

COMPÉTENCES

- Building Scalable, Self-Managed Platforms – Creating solutions that adapt and grow with the business.
- Solving Real-World Business Problems – Aligning technical solutions with company objectives.
- Improving Developer Efficiency – Reducing friction through automation, better tooling, and optimized workflows.
- Mentoring and Team Leadership – Supporting engineers and promoting best practices.

FOUNDER

FEEN — AI ACCOUNTING PLATFORM

HTTPS://WWW.FEEN.BE

Feen is an AI-powered accounting assistant for Belgian SMEs, helping entrepreneurs manage invoices, transactions, and financial insights in a single platform.
The platform automates financial administration by combining document extraction, reconciliation, dashboards, and accounting collaboration tools.

- Designed and built the full SaaS architecture
- Developed AI-powered financial document extraction
- Implemented automatic transaction reconciliation
- Built dashboards providing real-time financial insights
- Integrated Peppol readiness and company verification tools
- Developed scalable backend APIs and data pipelines
- Implemented LLM workflows for data processing and automation

Python • TypeScript • Next.js • AI/LLM APIs • Vector databases • Microservices • Cloud infrastructure

SYNQ — AI AGENT ORCHESTRATION PLATFORM

HTTPS://SYNQ.PIXLDEV.BE

Synq is a workflow orchestration platform for AI-native software delivery, acting as the control plane between engineering teams and AI agents.
The platform transforms unstructured AI usage into governed production pipelines, enabling multiple AI agents to collaborate safely across development workflows.

- DAG-based workflow orchestration for AI-driven development pipelines
- Parallel AI agent execution (backend, frontend, QA, DevOps, security)
- Quality gates with automated testing and validation before deployment
- Sandboxed execution environments for safe experimentation
- Role-based permissions and governance for AI-generated code
- Cost intelligence tracking token usage and compute per agent
- Failure detection with automated recovery and rollback
- Plugin architecture supporting multiple AI providers (GPT, Claude, Codex, custom models)

- Built orchestration engine coordinating multi-agent AI workflows
- Designed event-driven pipeline system from prompt → production
- Implemented streaming visibility into agent reasoning and execution
- Developed plugin system for integrating external tools and AI providers
- Built infrastructure for AI cost tracking and optimization

TECHNICAL SKILLS

SOFTWARE ARCHITECTURE & BACKEND DEVELOPMENT

LANGUAGES

TypeScript • JavaScript • Python

FRAMEWORKS

NestJS • Node.js • gRPC • REST • GraphQL

ARCHITECTURE

Microservices • Event-driven systems • Distributed systems • API-driven platforms

SCALABILITY

High-availability systems • queue processing • service orchestration • financial-grade reliability

DEVOPS

Azure • GCP • CI/CD • AKS / Docker

AI ENGINEERING

- **Backend Development:** Built and deployed tools leveraging context improvement, memory, and prompt engineering to optimize LLM calls and reduce token usage.
- **Infrastructure & Cloud:** Integrated vector databases and retrieval-augmented generation (RAG) patterns for enhanced information retrieval and semantic search.
- **Automation & Performance:** Automated deployment pipelines and ensured robust monitoring, error handling, and scalability for AI-powered services.

FRONTEND DEVELOPMENT

- **Modern frameworks:** Proficient in React.js, Next.js, and TypeScript for building fast, scalable, and SEO-friendly applications in headless CMS environments
- **System Design, UI/UX:** Skilled in transforming UI/UX designs into pixel-perfect, responsive interfaces using TailwindCSS, and JavaScript and tools like Figma.
- **Performance & accessibility:** Focused on web performance optimization (lazy loading, SSR/ISR), and accessibility (WCAG, ARIA, semantic HTML) for inclusive, high-ranking web experiences.

BUSINESS-CRITICAL SYSTEMS

- **FinTech & InsurTech Solutions:** Payments, pricing engines, compliance automation
- **Third-Party Integrations:** Banking APIs, insurance providers, external services
- **Contract & Policy Management:** Automated contract generation, bordereau reporting
- **Scalability & Optimization:** Data migration, performance tuning, distributed systems, dev tools, auto generation code

LEADERSHIP & TEAM GROWTH

- **Process Improvement:** Reducing inefficiencies in development workflows
- **Mentorship & Knowledge Sharing:** Supporting and guiding teams for better productivity
- **Cross-Team Collaboration:** Working with product, operations, and leadership

EXPERIENCE

BELFIUS, BRUSSELS — AI SYSTEMS ENGINEER

JULY 2025 - ONGOING

At Belfius, I build AI systems that work within the strict constraints of a regulated bank. This means
taking experimental AI technology and making it production-ready, secure, and compliant. I work
closely with Architecture teams to ensure everything we build meets banking standards.

● Designed retrieval systems that let LLMs access bank knowledge safely, using vector
databases and hybrid search
● Set up Azure infrastructure with proper security controls—managed identities, encryption,
zero-trust policies.
● Created FastAPI services that connect AI models to enterprise data while maintaining strict
access controls.
● Built AI agents that help with internal operations—automating tasks that previously required
manual work.

My role focuses on bridging the gap between cutting-edge AI technology and enterprise-grade
security requirements, ensuring every system is production-ready and meets regulatory standards.

COHABS, BRUXELLES — SENIOR SOFTWARE ENGINEER

AVRIL 2025 - JUIN 2025

At Cohabs, I focus on building a scalable payments service to focus on all edges cases, to help the team work more efficiently. And most important mirroring transactions to financial tool used by finance department, accounting & payment consolidation.

- Third party api integration for communication to accounting tool.
- Enhance workflow on events with error managing and queue process.
- Handle migration between growing services and changes.
- Focused on scalability, process automation, and system reliability in a growing platform.
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My role is not just about building software but helping the team deliver with more reliability and tracing between departments.

GREENOMY, BRUXELLES — SENIOR SOFTWARE ENGINEER

JAN 2024 - AVRIL 2025

At Greenomy, I focus on building a scalable SaaS platform while improving engineering workflows to help the team work more efficiently.

- Automated CI/CD processes, improving deployment speed and reliability.
- Introduced better development workflows, monitoring tools, and debugging strategies.
- Mentored engineers, reducing friction in problem-solving and improving overall productivity.
- Create features based on business needs and auto scalable solutions.
- Focused on scalability, process automation, and system reliability in a growing platform.

My role is not just about building software but helping the team deliver faster, more efficiently, and with better long-term stability.

QOVER, BRUXELLES — SENIOR SOFTWARE ENGINEER

2017 - 2024

I was part of the early technical team at Qover, helping transform it from a 13-person startup to a 150+ employee scale-up. I worked across all major systems, ensuring that the platform was scalable, automated, and built for long-term success.

- Built authentication, third-party integrations, pricing models, claims automation, and compliance systems.
- Developed whitelabel insurance solutions, making it easy to onboard and launch new partners at scale.
- Created contract automation, bordereau generation, financial reconciliation, and notification workflows.
- Helped migrate and optimize cloud infrastructure from AWS to GCP, ensuring high availability.
- Played a key role in building the foundation of a platform that is still running successfully at scale today.

Qover was a true end-to-end experience in building a high-growth Insurance as a service and embedded platform, from startup launch to full-scale enterprise operations.

EMAKINA, BRUXELLES — INTERNSHIP

2017

- Developed a digital signage platform for real-time content display across multiple screens.
- Integrated live data feeds (weather, schedules, news updates, etc.).
- Optimized secure authentication and data access processes.

FORMATION

INSTITUT PAUL LAMBIN, BRUXELLES — BACHELOR IN COMPUTER SCIENCE

2014 - 2017